

AI LITERACY FOR SOCIAL SCIENTISTS · SPECIAL LECTURE 01

사회과학자를 위한 AI Literacy 특강 시리즈

Rethinking the Research Process: How AI Is Changing Research for Students and Professors

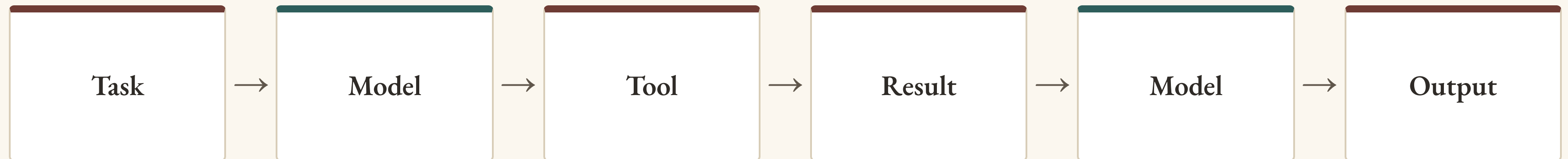
Steven Denney

Leiden University

Korea University · Friday 18 September 2026 · Moderator: Sung Eun Kim

A source I had cited twice did not say what I claimed. A machine found it.

The tools changed in kind, not degree. They read your files, run your code, and come back.



A chat window answers. An agent acts, sees what happened, and decides the next step, until the task is done or you stop it.

The question is not "AI or no AI." It is where to automate, what to protect, and where to collaborate.

Automate · Protect · Collaborate

Research and teaching now sort into three spaces, and each has its own rule.

Assembly line

AUTOMATE

Doing it adds nothing.

Apprenticeship

PROTECT

Doing it builds expertise.

Workbench

COLLABORATE

The machine changes the scale.

The map for the rest of the talk. Part one walks the three spaces. Part two stays at the workbench.

PART ONE

Allocating cognitive work

automate, protect, collaborate

Assembly line

AUTOMATE

Doing it adds nothing.

Apprenticeship

PROTECT

Doing it builds expertise.

Workbench

COLLABORATE

The machine changes the scale.

Automate the work that gains nothing from your attention.

Planning and tracking	An append-only handoff file records every session's decisions, so nothing lives only in an agent's memory.
Naming and filing	A drive of 14,238 scanned files became 438 catalogued volumes under one naming scheme.
Clerical	662 dictations, 54,796 words in one month, each cleaned by a model, for about eight dollars.
Formatting	One Markdown feedback letter builds to PDF and Word in a single command.
Conversion	Every PDF a project cites becomes a Markdown file that the checks can read.
Routine checks	Citations and numbers are verified at every commit, before any model reads a word.

**The claim is not that the machine can do these.
It is that they are poor uses of scarce attention.**

Protect the work through which expertise is built.

Reading	The source, not the summary. A summary cannot tell you what it left out.
First ideas	The question and the hunch, before any model has framed them for you.
Learning a procedure	Running the estimator by hand once, so you can tell when it is wrong later.
Some writing	The first draft of an argument. Originating a claim is a different skill from revising one a machine drafted.
Some coding	Enough to read a change to your code and know what it did to your data.
In-class assessment	The exam, the presentation, the seminar answer given without a prompt window.
Oral reasoning	Defending a claim in real time, which no text model can do for you.

Whether the machine can do the task is the wrong test.

The question is not whether the machine can do the task. It is whether you must do it to become able to judge the machine.

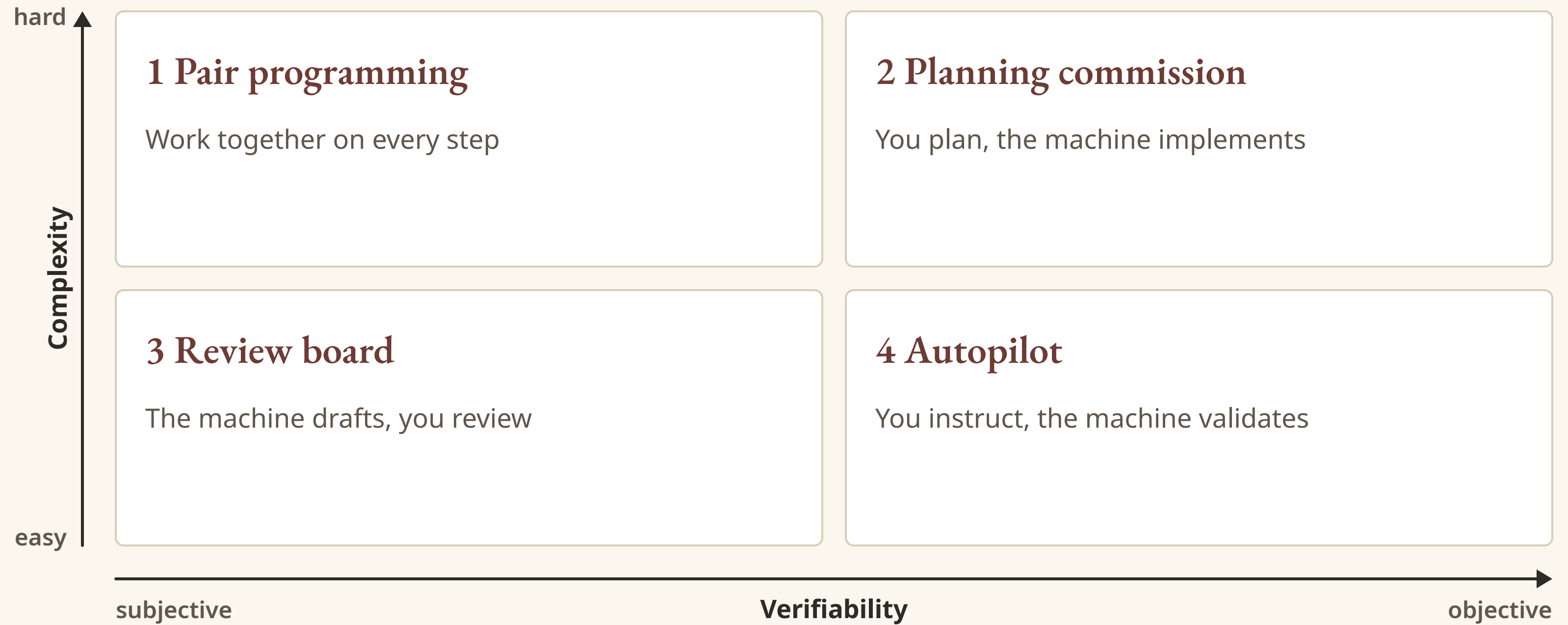
What I can offload after twenty years is what a student still has to struggle through.

"Human in the loop" says nothing. Say who holds epistemic responsibility.

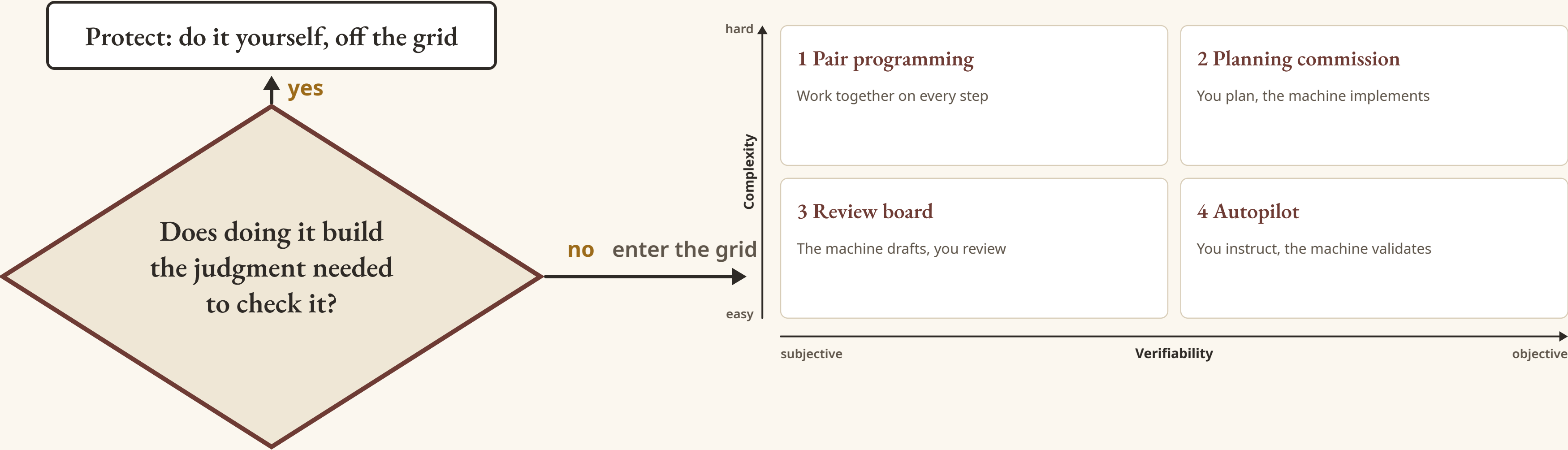
COLLABORATION, DEFINED

The human retains epistemic responsibility while the machine changes the scale, speed, or form of what can be done.

Inside the workbook, the working mode follows complexity and verifiability.



Before the grid, a gate: does doing it build the judgment needed to check it?



Kenny's grid sorts how to delegate. The gate asks whether you should.

Research raises the bar. The author certifies correctness.

Software

Goal: a product that works.

Mistakes: usually patched in the next release.

Understanding: no one person needs all of it.

Research

Goal: to know why or how something works.

Mistakes: found before publication, not after.

Understanding: the authors vouch for all of it.

So the workbench needs an audit trail.

PART TWO

At the workbench

two demonstrations and a coda

Assembly line

AUTOMATE

Doing it adds nothing.

Apprenticeship

PROTECT

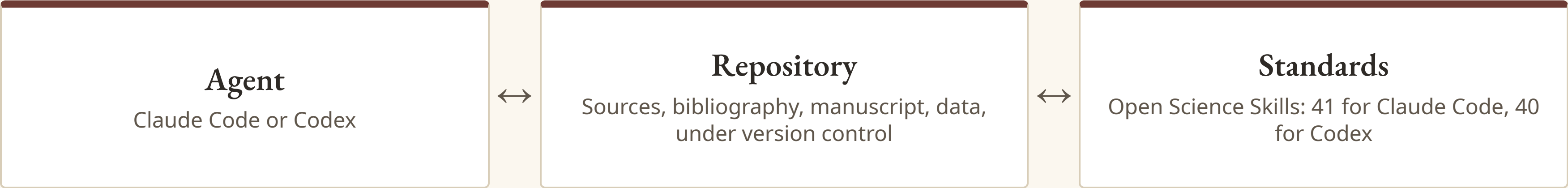
Doing it builds expertise.

Workbench

COLLABORATE

The machine changes the scale.

The workbench is an agent, a repository, and written standards.



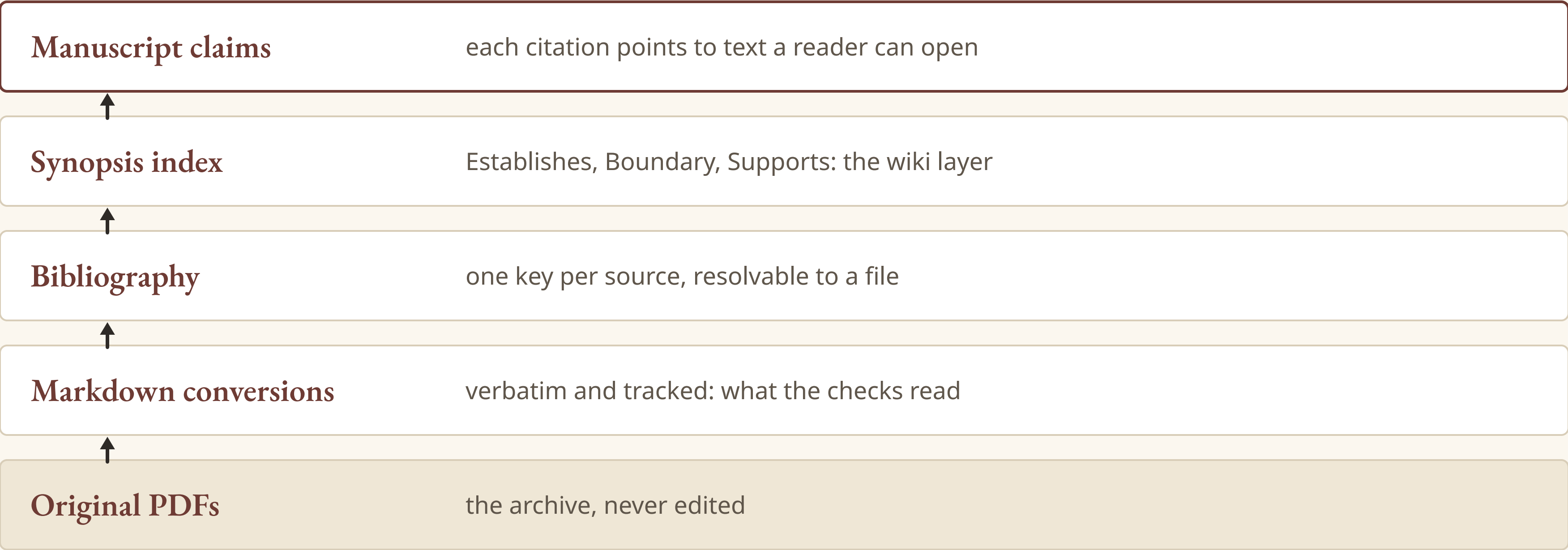
A skill encodes how the field thinks, not just what to do. It is where the standard lives.

Research runs on objects that do not talk to each other.



Each is real. None can vouch for the others.

A knowledge base connects them. Under each citation sits the source text, or a marked synopsis where the text is not held.



The synopsis layer follows Karpathy 2026's LLM wiki (raw sources, model-written pages, a periodic lint), as in Kenny 2026's demonstration. Here it sits on the verbatim layer, never in its place.

Ask the corpus, not the model.

› What do my sources say about ethnic and civic conceptions of the nation?

Kohn 1955 gives the classic statement of the dichotomy: membership by political principle versus by descent. `synopsis · SOURCES_SYNOPSES.md §1`

Tamir 2019 argues the boundary is blurred and that "civic" often exempts a nationalism from scrutiny. `full text held`

Kohn 1955 · one entry in the synopsis index

ESTABLISHES	Originates the influential dichotomy between a Western, rational, voluntarist-political (civic) nationalism and an Eastern, organic, cultural-ethnic nationalism.
BOUNDARY	A foundational historical-typological essay. The West/East geography of the binary is widely critiqued as overdrawn.
SUPPORTS	The civic conception centered on shared political principles and citizenship.

Now check whether the manuscript says what the source says.

> /oss:fact-check manuscript.md

✓ **SUPPORTED** "a modest, short-lived increase in youth turnout that fades within two election cycles"
Source, lindqvist2019: "modest, on the order of three to four points, and it does not survive beyond the second election."

~ **PARTIAL** "Higher social trust **causes** higher turnout"
Source, osei2020: "We document an association, not a mechanism." Suggested: *is associated with*.

✗ **CONTRADICTED** "compulsory voting also raises citizens' political knowledge"
Source, ferreira-nair2021: "Whatever compulsory voting does, it does not appear to teach."

The checks run in a chain, and the cheap deterministic ones run first.



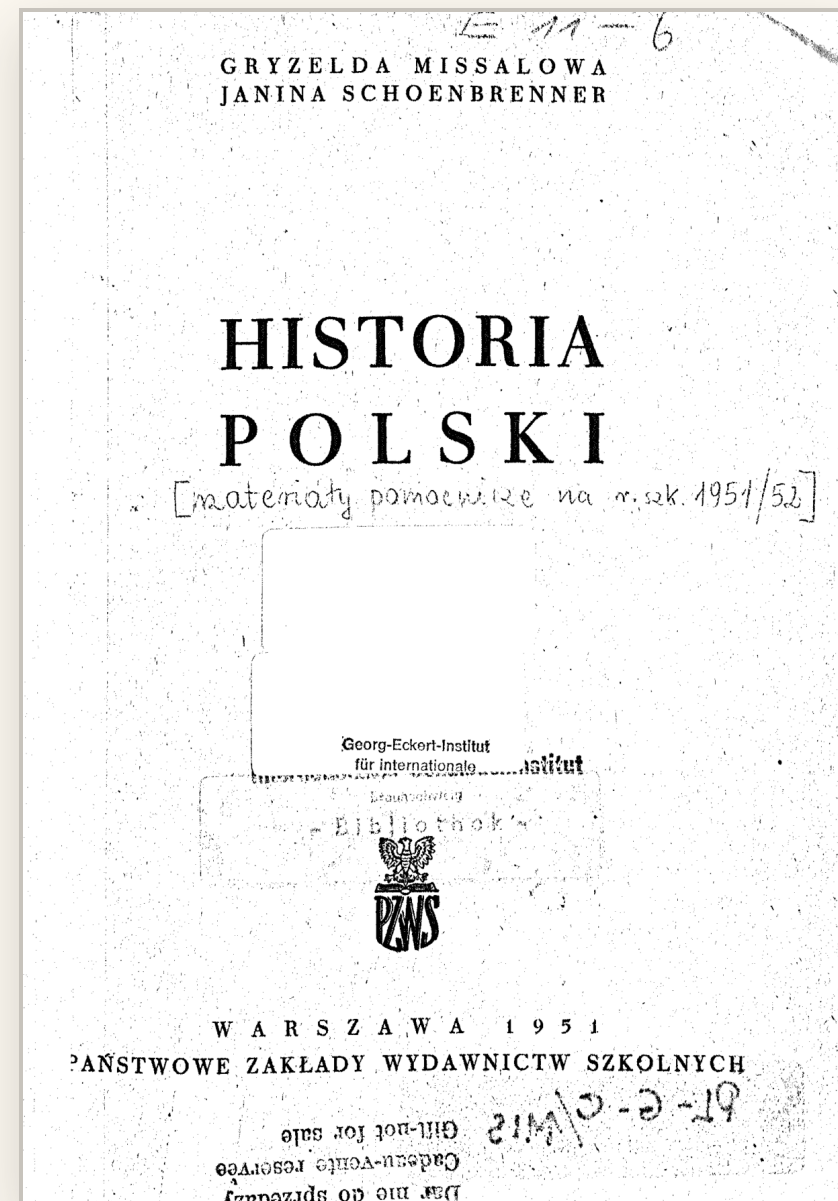
Scripts first, at every commit. The model reads last. Whatever was not checked is reported as NOT CHECKED.

Most of the talk about generative AI is about generation. The useful direction runs the other way.

I am increasingly interested in the opposite direction: using these systems to inspect, retrieve, verify, and challenge what I have written.

The machine is not producing prose. It is keeping an audit trail.

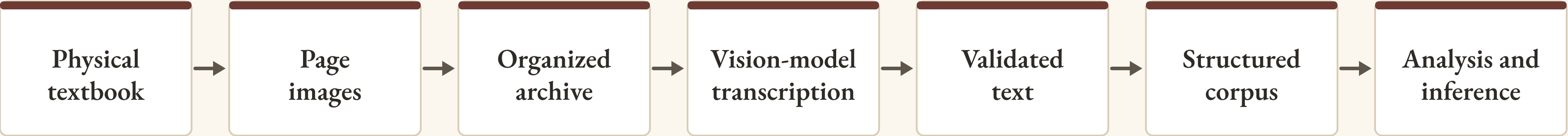
The knowledge base begins from documents that already exist. What if the material is not yet data?



*Historia Polski, 1951. Georg Eckert Institute library,
Braunschweig.*

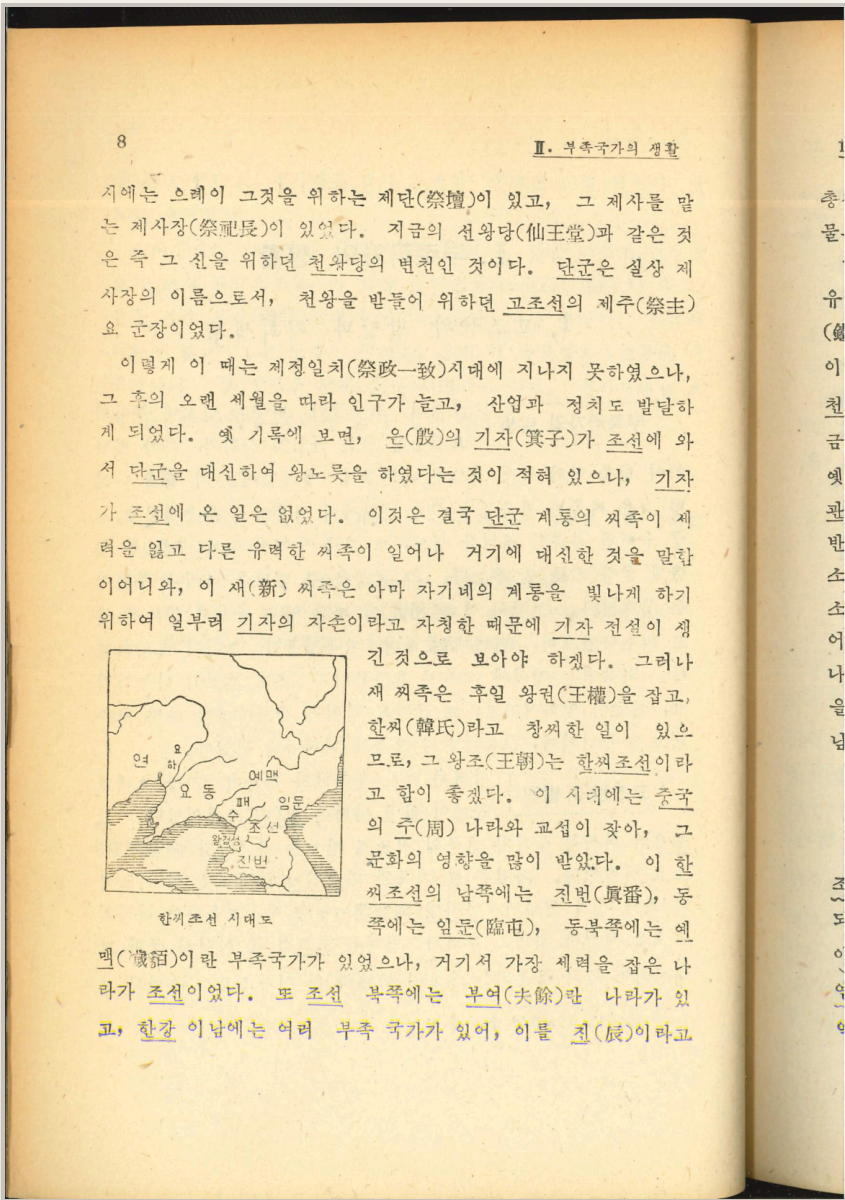
438 scanned volumes, 24 gigabytes of
page images, and not one searchable word.

Seven steps turn a textbook into data. The machine changed the cost of each.



438 volumes scanned and catalogued · 98,295 pages · 1948 to 2020 · 24 GB · transcription under way

On this page a vision-language model reads hangul and hanja whole. Tesseract without a Korean pack returns noise.



Vision-language model (Claude Sonnet 4.6, comparison run)

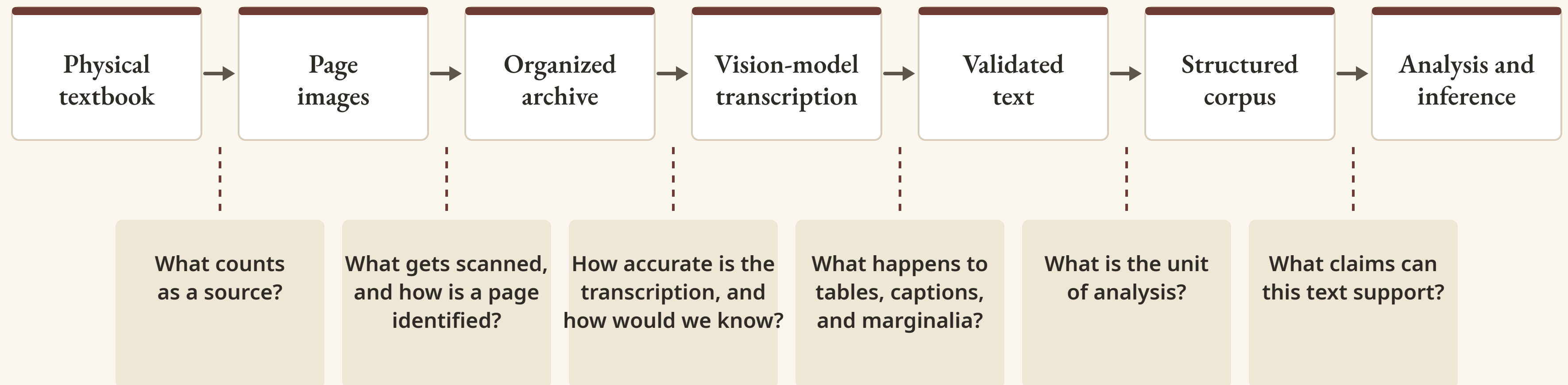
단군은 실상 제사장의 이름으로서, 천왕을 받들어 위하던 고조선의 제주(祭主)요 군장이었다.

Optical character recognition (Tesseract 5, English model only)

AAAs 2H] ade Ate ACM an, 2 ate ... no usable output

A 1956 history textbook, the page on Dangun and Gojoseon.
Same scan, same page.

Every arrow is a methodological decision.



This is research design, not assistance.

Without ground truth we cannot measure accuracy.
We can measure disagreement, and it is a warning.



When transcription error tracks language, it can pass for a cross-national difference in content.

Character-level disagreement among the six best-agreeing systems on body-text pages, from a 64-page, nine-system comparison. Preliminary. No human ground truth exists for this corpus, by a recorded decision.

The technology changes what can become data.

67

Korean history textbooks in a separate corpus, already machine-readable

11.3M

characters of cleaned text

3 of 4

open models must agree before a term counts

This corpus exists because transcription became cheap. The 438 volumes will grow it once their transcription is done.

The identity terms in the textbooks do not shift detectably at democratization in 1987.

THE PATTERN

Their character follows regime type and their timing follows curriculum reform, in 1992 to 1997 and again in 2002.

Sixty-seven textbooks. A term counts only when three of four open models agree. Observational, not causal.

One demonstration organizes what we know. The other makes new material knowable.

WHAT DO WE KNOW?	WHAT CAN WE MAKE KNOWABLE?
Organizing existing knowledge	Producing new research data
Literature and sources	Primary historical material
Retrieval	Extraction
Verification	Transformation
Claim checking	Corpus construction

Automate the machinery, not the judgment.

The same logic runs the supervision desk.



The judgment stays mine. The machinery around it is increasingly automated.

The allocation differs for students and for professors.

Students

Automate the clerical.

Protect reading, first ideas, and the procedures you are still learning.

Collaborate only where you can check the output.

Professors

Automate more of your machinery, and leave a trace.

Protect the formation of your students' judgment.

Collaborate: redesign the process around what is now feasible.

**The question for universities
is allocation, not adoption.**

Which work we keep, which we hand over, and where new combinations make the impractical practical.

Automate what does not deserve our attention.

Protect the work through which expertise is built.

Collaborate where machines expand what researchers can do.

Automate the machinery, not the judgment.

The skills, the site, and the methods are public.

SKILLS

Open Science Skills: 41 for Claude Code, 40 for Codex.

github.com/scdenney/open-science-skills

SITE

AI for Research: setup, demos, lectures, this deck.

scdenney.github.io/ai-for-research

SUBSTACK

Pixels and Patterns: the corpus methods, with code.

pixelsandpatterns.substack.com

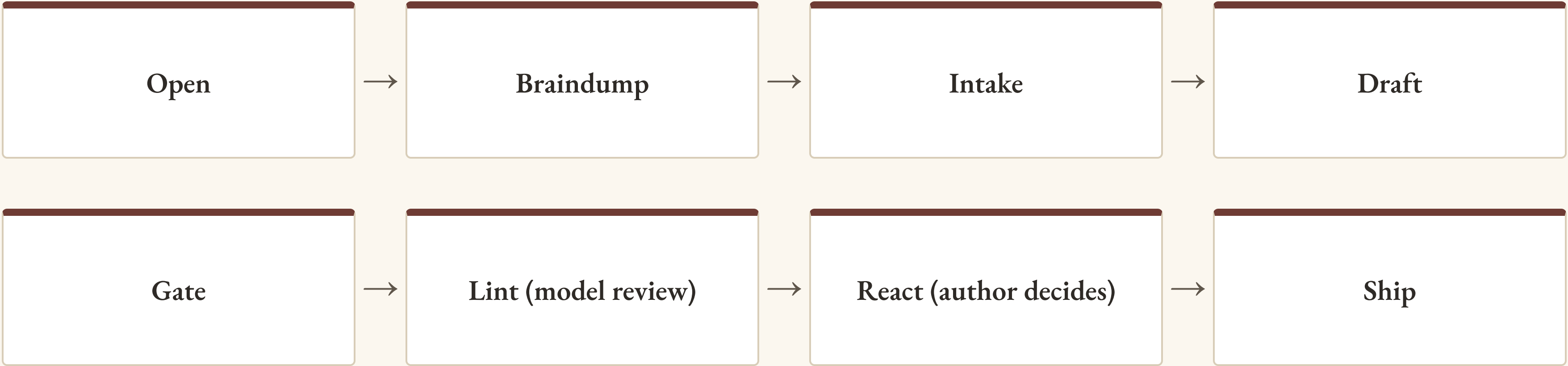


Appendix. Nine OCR systems were compared on the same 64 pages.

MODEL	PARAMETERS	PRECISION	SECONDS PER PAGE
Qwen3.5-35B-A3B	35B, 3B active	GPTQ-Int4	6.2
Qwen3-VL-32B	32B	NF4	30.0
Qwen3.5-9B	9B	NF4	22.8
Gemma 4 E4B	4.5B effective	BF16	15.4
MiniCPM-V 4.5	8B	BF16	53.3
DeepSeek-OCR-2	3.4B	BF16	12.1
GPT-4.1	API	n/a	8.2
Claude Sonnet 4.6	API	n/a	17.3
Tesseract	CPU	n/a	1.7

Seconds per page. Local models ran on one NVIDIA A40, Tesseract on CPU, and the two APIs over the network. Only Qwen3.5-35B ran under vLLM. Run of 7 April 2026, preliminary.

Appendix. The deliverable pipeline runs in eight steps.



One wiki per piece of research, one manifest per deliverable. The gate is a script that runs at every commit. Lint is a model review that runs only after the gate passes. This deck was built through it.

Appendix. Two commands install the skills.

```
claude plugin marketplace add scdenney/open-science-skills
```

```
claude plugin install oss@open-science-skills
```

Then name a skill, such as /oss:fact-check, or describe the task and let the matching skill load.

Appendix. Everything cited on these slides, in full.

Kenny 2026	Christopher T. Kenny, "Agentic AI for Political Science Research," Center for the Study of Democratic Politics, Princeton, 10 September 2026. github.com/christopherkenny/skills
Swaney 2026	Colin Swaney, "Claude Overload," Data-Driven Social Science workshop, Princeton, 20 March 2026. github.com/princeton-ddss/claude-overload
Karpathy 2026	Andrej Karpathy, "LLM Wiki," April 2026. gist.github.com/karpathy/442a6bf555914893e9891c11519de94f
Kremer 1993	Michael Kremer, "The O-Ring Theory of Economic Development," Quarterly Journal of Economics 108(3): 551–575.
Paglayan 2026	Agustina S. Paglayan, "Education as a Political Institution," Annual Review of Political Science 29: 525–546.
Denney 2026	Steven Denney, "From Pixels to Patterns: Vision-Language OCR and LLM-Based Text Analysis," Machine Collaborators, 25 June 2026.
Denney and van de Pol 2026	Steven Denney and Aron van de Pol, "Constructing the Nation," working paper on Korean history textbooks.